

Quantum Teleportation Quick Reference Sheet

What is Quantum Teleportation?

- A protocol that transmits a qubit state from sender (Alice) to receiver (Bob)
- Requires: 1 shared entangled pair (e-bit) + 2 classical bits of communication
- Teleports quantum information, not matter
- Preserves all quantum properties, including entanglement with other systems

Prerequisites

- Alice and Bob share an entangled Bell pair in state $|\phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$
 - Alice has qubit A
 - Bob has qubit B
- Alice has qubit Q in unknown state $\alpha|0\rangle + \beta|1\rangle$ that she wants to teleport

Protocol Steps

1. Alice's Operations:

- Apply CNOT gate with Q as control, A as target
- Apply Hadamard gate to Q
- Measure both A and Q in the standard basis
- Send the two classical bits (a and b) to Bob

2. Bob's Operations:

- Apply corrections based on the classical bits received:
 - If $a = 0, b = 0$: Apply I (do nothing)
 - If $a = 0, b = 1$: Apply Z gate
 - If $a = 1, b = 0$: Apply X gate
 - If $a = 1, b = 1$: Apply ZX gates

3. Result:

- Bob's qubit B is now in state $\alpha|0\rangle + \beta|1\rangle$ (the original state of Q)
- Original qubit Q has been measured, destroying its state
- The entangled pair is "consumed" in the process

Key Properties

1. No faster-than-light communication

- Classical bits must be transmitted
- Each measurement outcome occurs with exactly $\frac{1}{4}$ probability regardless of input state

2. No cloning

- Original state of Q is destroyed
- Only one copy exists at the end (with Bob)

3. Resource consumption

- Each teleportation consumes one e-bit
- Requires two bits of classical communication

4. Perfect fidelity

- Works for any qubit state
- Maintains all quantum correlations with other systems

Mathematical Analysis

1. Initial state:

$$|\pi_0\rangle = |\phi^+\rangle \otimes (\alpha|0\rangle + \beta|1\rangle) \quad (1)$$

$$= \frac{|00\rangle + |11\rangle}{\sqrt{2}} \otimes (\alpha|0\rangle + \beta|1\rangle) \quad (2)$$

$$= \frac{\alpha|000\rangle + \alpha|110\rangle + \beta|001\rangle + \beta|111\rangle}{\sqrt{2}} \quad (3)$$

2. After CNOT:

$$|\pi_1\rangle = \frac{\alpha|000\rangle + \alpha|110\rangle + \beta|011\rangle + \beta|101\rangle}{\sqrt{2}} \quad (4)$$

3. After Hadamard:

$$|\pi_2\rangle = \frac{1}{2}(\alpha|0\rangle + \beta|1\rangle)|00\rangle + \frac{1}{2}(\alpha|0\rangle - \beta|1\rangle)|01\rangle \quad (5)$$

$$+ \frac{1}{2}(\beta|0\rangle + \alpha|1\rangle)|10\rangle + \frac{1}{2}(\beta|0\rangle - \alpha|1\rangle)|11\rangle \quad (6)$$

4. After measurement and corrections:

$$\text{Bob's qubit is always in state } \alpha|0\rangle + \beta|1\rangle \quad (7)$$

Measurement Outcomes and Corrections

Outcome	Probability	Bob's Qubit State	Correction
$a = 0, b = 0$	$\frac{1}{4}$	$\alpha 0\rangle + \beta 1\rangle$	None (I)
$a = 0, b = 1$	$\frac{1}{4}$	$\alpha 0\rangle - \beta 1\rangle$	Z
$a = 1, b = 0$	$\frac{1}{4}$	$\beta 0\rangle + \alpha 1\rangle$	X
$a = 1, b = 1$	$\frac{1}{4}$	$\beta 0\rangle - \alpha 1\rangle$	ZX

Teaching Tips

- Emphasize that no quantum information is extracted during the process
- Demonstrate with specific examples (e.g., $|+\rangle$ state or $|0\rangle$ state)
- Use the circuit as a visualization aid while explaining each step
- Relate to EPR paradox and Bell's theorem for advanced students
- Show the case where qubit Q is entangled with another system (labeled R) for a complete understanding